frequencies between -0.3 and -3 kHz, respectively. Even though the negative frequencies in the original audio signal are somewhat elusive and abstract, the resulting frequencies in the lower sideband are as real as you could want them to be. The ghosts have taken human form!

Communication engineers live and die by this type of frequency domain analysis. For example, consider the frequency spectrum for television transmission. A standard TV signal has a frequency spectrum from DC to 6 MHz. By using these frequency shifting techniques, 82 of these 6 MHz wide channels are stacked end-to-end. For instance, channel 3 is from 60 to 66 MHz, channel 4 is from 66 to 72 MHz, channel 83 is from 884 to 890 MHz, etc. The television receiver moves the desired channel back to the DC to 6 MHz band for display on the screen. This scheme is called **frequency domain multiplexing**.

The Discrete Time Fourier Transform

The Discrete Time Fourier Transform (DTFT) is the member of the Fourier transform family that operates on *aperiodic*, *discrete* signals. The best way to understand the DTFT is how it relates to the DFT. To start, imagine that you acquire an N sample signal, and want to find its frequency spectrum. By using the DFT, the signal can be decomposed into $N/2 + 1$ sine and cosine waves, with frequencies equally spaced between zero and one-half of the sampling rate. As discussed in the last chapter, padding the time domain signal with zeros makes the period of the time domain *longer*, as well as making the spacing between samples in the frequency domain *narrower*. As N approaches infinity, the time domain becomes *aperiodic*, and the frequency domain becomes a *continuous* signal. This is the DTFT, the Fourier transform that relates an *aperiodic*, *discrete* signal, with a *periodic*, *continuous* frequency spectrum.

The mathematics of the DTFT can be understood by starting with the synthesis and analysis equations for the DFT (Eqs. 8-2, 8-3 and 8-4), and taking N to infinity:

EQUATION 10-1
The DTFT analysis equation. In this relation, $x[n]$ is the time domain signal with n running from 0 to $N-1$. The frequency spectrum is held in: $ReX(\omega)$ and $ImX(\omega)$, with ω taking on values

$$ReX(\omega) = \sum_{n=-\infty}^{+\infty} x[n] \cos(\omega n)$$

$$ImX(\omega) = -\sum_{n=-\infty}^{+\infty} x[n] \sin(\omega n)$$

EQUATION 10-2
The DTFT synthesis equation.

$$x[n] = \frac{1}{\pi} \int_{0}^{\pi} ReX(\omega) \cos(\omega n) - ImX(\omega) \sin(\omega n)\, d\omega$$